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# Configurational effects of personal innovativeness, self-efficacy, and perceived risk on AI adoption in media students

Yanling Lan, Sihang Liu, Hao Chen✉ & Linjie Xia

**Generative artificial intelligence (G-AI) is driving systemic transformations in the media industry, redefining the cultivation paradigm of media majors in higher education. This shift necessitates that future media professionals integrate AI thinking and master the application of digital intelligence technologies. To address the knowledge gap in understanding media students' adoption of AI technologies, this study extends the classical Technology Acceptance Model by incorporating personal innovativeness, AI self-efficacy, and perceived risk, thereby constructing a new and comprehensive model. A survey of 588 media students was conducted, and the data were analyzed using PLS-SEM and fsQCA. The results indicate that: (1) personal innovativeness and AI self-efficacy significantly enhance both perceived usefulness and perceived ease of use; (2) perceived usefulness and perceived ease of use are core determinants of AI behavioral intention, mediating the effects of personal innovativeness and AI self-efficacy; (3) perceived risks can negatively moderate the effect of perceived usefulness and perceived ease of use on AI behavioral intention; (4) AI behavioral intention is shaped not by a single linear determinant but by intersecting configurations of factors, leading to three distinct enhancement pathways: self-driven, efficacy-oriented, and risk-resistant. This study advances the Technology Acceptance Model by integrating psychological and risk-based constructs, offering theoretical insights into the multidimensional driving forces of AI adoption and practical strategies for fostering AI competencies in media students.**

**Keywords** Media education, Artificial intelligence(AI), Technology acceptance model (TAM)

The human-machine collaborative production model introduced by artificial intelligence (AI) in the media has fundamentally transformed the industry's ecosystem, reshaping the media's stance on professional knowledge and competencies (Moller and Skovsgaard et al., 2024)[1]. AI technology is rapidly and continuously altering the media organizational structure, business workflows, operational practices, and the roles and skills of media professionals, while also adapting to the evolving demands of the knowledge literacy and technical competencies of future media talent. Despite rapid technological advancements, media higher education continues to face substantial knowledge gaps and practical challenges. The debate over whether technological progress truly equates to meaningful advancement, alongside growing skepticism about "machines as non-human agents," exacerbates tensions within the field. Lingering concerns regarding technological transparency, creative accountability, and ethical considerations only serve to escalate these conflicts (Monti, 2019)[2]. Porlezza(2023)[3] highlights the widening cognitive divide within both the media industry and academia, fueling rising skepticism toward intelligent technologies. The surge in AI-generated misinformation (Whyte, 2020)[4], coupled with difficulties in source verification and ensuring content authenticity in AI-driven news (Gonzalez-Arias & Lopez-Garcia, 2023)[5], has led to a trust crisis and disengagement with AI. Furthermore, the exorbitant operational costs of AI systems, coupled with their lack of user-friendliness, interfaces present significant barriers to developing students' AI application skills (Stray, 2021[6]; Borchardt, 2022[7]. Meanwhile, the perspectives of media education faculty, curricula designs, and attitudes of students are trailing behind the requirements of the intelligent media. A study by Wenger (2024)[8]found that none of the 14 U.S. media education institutions surveyed had established a comprehensive AI education program. While the number of intelligent courses has increased, their content often emphasizes technological popularization, and the teaching effectiveness frequently falls short of expectations.

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Additionally, there is a lack of focus on the interaction between students, technology, and teachers. The approach overemphasizes the role of technology and the authority of educators, while underestimating the importance of students' intrinsic motivation and active participation.

Research has consistently highlighted that the willingness and participation of individuals are key determinants in enhancing the adoption of AI technologies among media professionals and students (Kuai, 2024[9]; Gomez-de-Agreda and Feijoo, 2021[10]. Brennen and Howard et al.(2020)[11] demonstrated, using the UK media industry as a case study, that technology expectations, AI self-efficacy, and AI adoption are closely interconnected. Gutierrez-Lopez and Porlezza et al.(2022)[12], through in-depth interviews with employees at two London newsrooms, identified that innovation awareness and practical orientation are crucial factors for AI integration in the media. Mitova and Blassnig et al.(2024)[13], in a comparative study across five countries (Netherlands, Poland, Switzerland, UK, and USA), confirmed that individual characteristics significantly influence AI's application in the media. Gil-Ruiz and Dominguez-Lloria (2025)[14] conducted a mixed-method study with 55 students from two Spanish universities and found that innovation capability, self-expression, and perceived technological efficacy are significant drivers of AI tool usage.

Previous studies have indicated that, apart from external environmental factors and curriculum design, students' personal characteristics, cognitive abilities, and willingness to embrace new technologies significantly influence their acceptance of AI. Meanwhile, within the context of higher education, there has been a growing number of empirical studies to explore how college students accept and use AI technology (Rodway and Schepman,2023[15]; Oc and Gonsalves et al.,2025[16], especially some scholars have expanded TAM by incorporating additional constructs such as AI literacy, AI self-efficacy, personal innovativeness, and perceived risk to predict students' intentions and behaviors when utilizing AI tools (Ma and Lei, 2024[17]; Al-Adwan and Li et al., 2023[18]. However, there are still some gaps within the field of media education. First, in terms of research content, it is crucial to develop a more comprehensive theoretical framework to construct a model that analyzes the influencing factors and mechanisms of AI technology adoption among media students. Second, in terms of methodology, more comprehensive empirical approaches should be utilized to conduct in-depth analyzes supported by robust data. Third, regarding research subjects, current studies on AI education in the media field primarily focus on industry professionals, while insufficient attention is given to college students, who represent the future talent pool of the media sector.

Building on the above, this study aims to address three core issues. (1) Extend the classical TAM by incorporating key psychological factors, including personal innovativeness, AI self-efficacy, and perceived risk, to establish a more multidimensional theoretical framework for analyzing how media students adopt AI technologies. (2) Combine Partial Least Squares Structural Equation Modeling (PLS-SEM) with Fuzzy-Set Qualitative Comparative Analysis (fsQCA). This complementary methodological approach not only enables path analysis of relationships among variables, but also reveals how different combinations of factors collectively influence AI adoption intentions. (3) Focus on media students and explore their intentions to adopt AI technologies within the context of higher education, thereby addressing the issue that there have been relatively few studies. In summary, this research contributes by developing a comprehensive theoretical framework and providing empirical evidence on the determinants of AI adoption among media students. The findings offer both theoretical insights into the integration of psychological factors within the TAM and practical guidance for optimizing AI-related curriculum design in higher education, with the broader aim of enhancing the preparation and training of intelligent media professionals.

## Theoretical development and research hypotheses

### Technology acceptance model and its application in higher education

The TAM, derived from the Theory of Reasoned Action, posits that an individual's technology adoption behavior is influenced by their willingness to use, which is shaped by personal subjective norms and attitudes (Douglass, 1977)[19]. David (1989)[20] formalized TAM's framework to explain how individuals, particularly in work settings, adopt and use new technologies, with core variables including perceived usefulness, perceived ease of use, and attitudes toward using the technology, behavioral intention, and actual usage. Subsequent research by Davis and colleagues emphasized that "behavioral intention" should replace "overall attitude" as the central element, providing a more precise explanation of user behavior (Venkatesh and Davis, 1996)[21]. In 2000, Venkatesh (2003)[22] introduced TAM2, which added subjective norm and cognitive workload to address environmental and cognitive factors in technology adoption. In 2003, the Unified Theory of Acceptance and Use of Technology (UTAUT) emerged, incorporating TAM's core variables alongside four additional dimensions: performance expectancy, effort expectancy, social influence, and facilitating conditions. TAM remains a foundational theory for understanding individual technology acceptance and usage (Cai, 2022)[23], with broad applicability across fields such as medicine, sociology, economics, communications, and education (Lin and Chou et al., 2016[24]; Fareed and Kirkil, 2025[25].

In recent years, the TAM has become a prominent framework for examining the adoption and use of AI technologies in higher education. Granić (2022)[26]conducted a systematic review and analysis of studies on technology acceptance and use in educational contexts, finding that the vast majority of quantitative research employed TAM as the guiding framework. Among its constructs, perceived ease of use (PEOU) and perceived usefulness (PU) have been consistently identified as the most significant predictors of technology adoption. For example, Zou and Huang (2023)[27]applied TAM to investigate doctoral students' acceptance of ChatGPT in academic writing, confirming that both PEOU and PU exert a significant positive influence on behavioral intention.

Although the TAM has been widely applied and validated in explaining the acceptance and use of intelligent technologies in higher education, empirical research specifically focusing on media education—a distinct subfield within higher education—remains relatively scarce. In recent years, only a limited number of studies,

such as that by Tahat and Habes (2023)[28], have employed TAM to construct theoretical frameworks and utilized structured questionnaires to examine mass communication undergraduates' attitudes toward, and willingness to adopt, information and intelligent technologies. Moreover, the integration of AI technologies into the media sector is still at an early stage, and within media education, the development of curricula related to intelligent technologies remains largely exploratory and experimental.

## The expansion of TAM in college students' application of AI

Compared with previous digital, information, and media technologies, artificial intelligence (AI) demonstrates distinctive features such as high complexity, low transparency, and rapid evolution. These characteristics indicate that its acceptance and application in higher education cannot be fully explained by the classic TAM, which mainly relies on "Perceived Usefulness" and "Perceived Ease of Use ", but rather require the integration of additional research dimensions and influencing factors. For example, Ma and Lei (2024)[17] investigated the factors affecting teachers' and students' acceptance of AI technology in the context of digital learning based on the TAM. On the basis of the original variables, they incorporated AI literacy, AI self-efficacy, and output quality, and confirmed that perceived usefulness, perceived ease of use, and AI self-efficacy were the most critical factors influencing behavioral intention to use AI. Similarly, Al-Adwan and Li et al. (2023)[18] employed TAM to predict university students' intention to use intelligent learning platforms and identified perceived usefulness, perceived ease of use, and personal innovativeness as the main facilitating factors. At the same time, AI self-efficacy and personal innovativeness were found to be the main determinants of perceived ease of use and perceived usefulness, whereas perceived risk served as a major inhibiting factor. Among these, personal innovativeness, AI self-efficacy, and perceived risk have been most frequently highlighted in previous studies as significant influencing variables.

Individuals with higher levels of personal innovativeness are typically eager to outperform their peers, hold more positive attitudes toward innovative technologies or services, and demonstrate stronger intentions to adopt them in practice. Chen and Li (2020)[29] further argued that, in the era of AI, individuals with high personal innovativeness also tend to become early adopters of intelligent technologies or emerging products. In the context of higher education, Lu and Liu (2008)[30] found that students with stronger innovative tendencies usually exhibit more optimistic attitudes toward technology and greater initiative in adopting it.

AI self-efficacy refers to an individual's belief in their ability to perform a specific task, regardless of potential difficulties, and this expectation can directly or indirectly influence their behavior (Williams and Rhodes et al., 2016)[31]. Montag and Kraus (2024)[32] noted that AI self-efficacy can positively influence individuals' trust in AI and further reduce their negative perceptions of it. Similarly, Kim and Kim (2024)[33] suggested that when individuals possess a high level of AI self-efficacy, they are more likely to generate stronger positive expectations during the process of technology use.

Perceived risk mainly refers to individuals' anticipation of potential adverse outcomes or uncertainties arising from the use of technology, including concerns about security, privacy protection, and reliability (Hwang and Kim,2024)[34]. Lee and Song (2013)[35] pointed out that perceived risk is the greatest challenge individuals face when adopting new technologies; when individuals perceive that technology may pose risks, they are more likely to conclude that it is not worth the attempt. In the field of AI, as an emerging technology, the absence of proper AI governance may render any AI-driven solutions a threat to actual use. Duffett and Zaharia et al. (2024)[36], using Romanian students as their research subjects, confirmed that perceived risk stemming from technological distrust significantly reduces students' willingness to use AI technologies.

## Hypothesis development

Building on existing research, this study advances the theoretical foundation to explain the mechanisms and determinants of individual AI adoption in media education. It integrates personal innovativeness, AI self-efficacy, and perceived risk into the TAM model, which can comprehensively explain how personality traits, ability perception, and risk assessment affect students' AI behavioral intention. These extensions enhance the TAM's explanatory depth and applicability(See Figure 1).

The Impact of Personal Innovativeness on the Acceptance of AI Technologies.

According to Rogers' diffusion of innovations theory, personal innovativeness is defined as the degree to which an individual adopts innovative or novel ideas(Rogers,2023)[37]. Based on the study by Lu and Liu (2008)[30], students with higher levels of innovativeness are more likely to perceive the usefulness and ease of use of technology in task performance (Van and Schepers,2008)[38]. Gunness and Matanda (2023)[39] also confirmed that, in learning environments, students' personal innovativeness has a significant positive effect on both perceived usefulness and perceived ease of use. Building on the existing literature, this study proposes the following hypotheses:

H1: Personal innovativeness has a significant positive effect on media students' perceived usefulness of AI technologies.

H2: Personal innovativeness has a significant positive effect on media students' perceived ease of use of AI technologies.

The Impact of AI Self-efficacy on the Acceptance of AI Technologies.

AI self-efficacy is defined as an individual's confidence, capability, and willingness to engage in learning through the use of AI tools (Wang and Chuang,2024)[40]. Kim and Kim (2024)[33]suggested that when individuals possess a high level of self-efficacy, they tend to develop stronger positive expectations during the process of technology use, which further enhances their perceived usefulness and perceived ease of use of AI technologies. Based on the existing literature, this study proposes the following hypotheses:

H3: AI self-efficacy has a significant positive effect on media students' perceived usefulness of AI technologies.

H4: AI self-efficacy has a significant positive effect on media students' perceived ease of use of AI technologies.

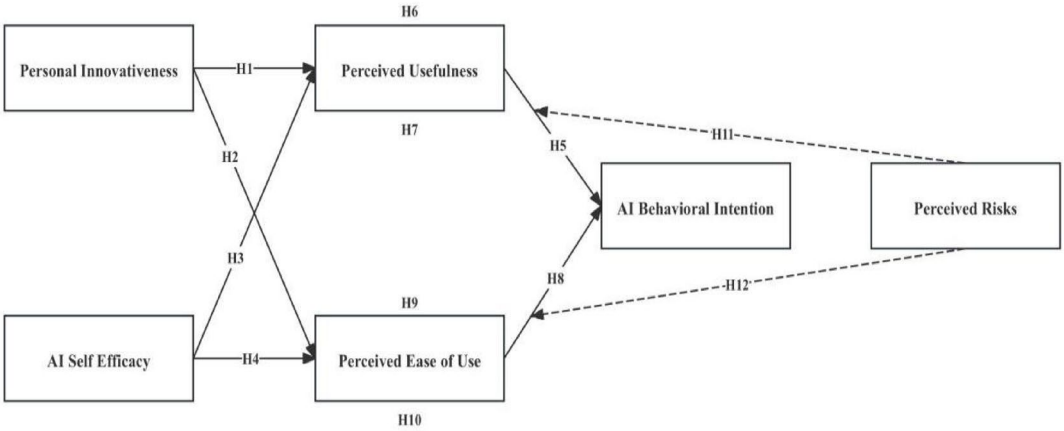


**Fig. 1**. Structural equation modeling framework based on the extended TAM.

The Impact of Perceived Usefulness on the Behavioral Intention of AI Technologies.

Perceived usefulness is a key factor directly influencing an individual's intention to continue using intelligent technologies (Mouakket, 2015)[41]. When individuals believe in the usefulness of a technology and this belief aligns with their values, they are more inclined to use it (Li, 2023)[42]. In higher education, studies by Algerafi and Zhou et al.(2023)[43] and Sanchez-Prieto and Izquierdo-Alvarez et al.(2025)[44] confirm that perceived usefulness significantly drives students' technology adoption, facilitating knowledge exploration. Ates and Gunduzalp (2025)[45] further show that individual traits such as AI self-efficacy, personal innovativeness, and subjective norms positively influence perceived usefulness, which in turn impacts behavioral intention. Based on these findings, the following hypothesis is proposed:

H5: Perceived usefulness has a significant, positive effect on media students' behavioral intention to use AI tools.

H6: Perceived usefulness mediates the relationship between personal innovativeness and media students' behavioral intention to use AI tools.

H7: Perceived usefulness mediates the relationship between AI self-efficacy and media students' behavioral intention to use AI tools.

The Influence of Perceived Ease of Use on the Behavioral Intention of AI Technologies.

Perceived ease of use pertains to users' anticipation of minimal obstacles in adopting a new technology, suggesting it can be readily grasped without intensive learning endeavors (Warshaw and Davis, 1985)[46]. Davis (1989)[20], in formally developing TAM, demonstrated that perceived ease of use shapes users' attitudes and intentions toward technology adoption—an effect further validated in user experience research. Empirical evidence consistently confirms TAM, highlighting perceived ease of use as a key determinant of emerging technology usage (Venkatesh and Davis, 2000)[47]. In higher education, studies have also identified perceived ease of use as a key factor predicting adoption. Zhao and Yan et al. (2024)[48]found that when students perceive AI tools as convenient, simple, and operable, they integrate them more frequently into learning tasks. Similarly, Ayanwale and Molefi (2024)[49]reported that perceiving intelligent tools as user-friendly strengthens students' future adoption intentions and fosters more positive usage attitudes. Drawing on these insights, the following hypothesis is proposed:

H8: Perceived ease of use significantly and positively influences media students' behavioral intention to use AI tools.

H9: Perceived ease of use mediates the relationship between personal innovativeness and media students' behavioral intention to use AI tools.

H10: Perceived ease of use mediates the relationship between AI self-efficacy and media students' behavioral intention to use AI tools.

The Moderating Effect of Perceived Risk.

Perceived risk primarily refers to individuals' anticipation of potential negative outcomes or uncertainties associated with the application of technology. Duffett and Zaharia et al. (2024)[36] conducted a study involving Romanian students, confirming that technological distrust significantly reduced students' willingness to use AI technology in practice. Based on existing related research, this study proposes the following hypothesis:

H11: Perceived risk negatively moderates the relationship between perceived usefulness and media students' behavioral intention to use AI tools.

H12: Perceived risk negatively moderates the relationship between perceived ease of use and media students' behavioral intention to use AI tools (Fig. 1).

## Research design and methodology

### Methods

This study integrates Smart-PLS (V4.1) and fsQCA methods. PLS-SEM is used for predictive exploratory factor analysis, effectively handling multidimensional data, forming measurement models, and analyzing relationships

between higher-order latent variables. fsQCA, a fuzzy-set qualitative comparative analysis, better elucidates causal and hierarchical relationships, offering strong applicability in both theoretical and practical contexts.

PLS-SEM is widely employed across various disciplines, including education, economics, and computer science, to evaluate large and complex models (Hair and Hult, et al., 2021)[50]. It follows a causal prediction paradigm, aligning with the objective of testing the predictive power of established theoretical models (Sarstedt and Hair, et al., 2022)[51]. PLS-SEM demonstrates strong predictive capabilities, with composite model characteristics that facilitate path estimation based on sample data and enable further predictive analysis through latent variable composites (Hair and Ringle, et al., 2019)[52]. Compared to other structural equation modeling techniques, PLS-SEM offers superior accuracy in both predictive ability and statistical performance, making it particularly well-suited for analyzing the effects of emerging technological factors (Donath and Holenko-Dlab et al., 2024)[53].

Is employed to effectively uncover the complex relationships between various causal combinations, particularly in analyzing individuals' willingness to use and adopt technology (Chanda and Vafaei-Zadeh et al.,2024)[54]. In recent years, it has found widespread application across diverse fields such as information systems, e-commerce, marketing, data science, education, and psychology (Greckhamer and Furnari et al.,2018)[55]. Pappas and Woodside (2021)[56] recommend combining fsQCA with other data analysis techniques to uncover different perspectives within the same dataset, thus enhancing the credibility and validity of the results. Liu and Chen, et al. (2025)[57] used a combination of SmartPLS and fsQCA to explain the complex interactions among factors driving user behavior within the context of AIGC. In their study, fsQCA effectively explored the causal relationships between multiple factors after examining both direct and indirect effects through PLS-SEM, and identified distinct pathways leading to users' behavioral intention to use the technology. The combined use of SmartPLS and fsQCA has proven to be a highly credible, feasible, and persuasive method for uncovering users' willingness to adopt emerging technologies and for improving adoption pathways (Li and Liu, et al., 2025[58]; Tang and Su, 2025[59]; Zhou and Wu, 2025[60].

Given that media students' behavioral intention to use AI is influenced by multiple factors, this study first employs PLS-SEM to assess the impact of various latent variables on behavioral intention. Building upon the structural equation model results, fsQCA is subsequently utilized to explore the combinatory effects of variables and identify specific pathways that shape media students' behavioral intention.

## Data collection

This study employs a questionnaire survey method, conducted offline among media students at universities in China. The questionnaire was meticulously crafted using well-established scales and indicators, which have been previously validated in research. These scales and indicators underwent rigorous reliability and validity assessments, as detailed in Table 1. The complete questionnaire is available in Appendix A. Data collection took place from February to June, 2025, yielding 610 completed questionnaires. After excluding invalid responses (e.g., completion times under one minute or incomplete answers), After ensuring the validity and security of the data, 588 valid questionnaires were retained for comprehensive analysis.

## Statement

This study confirms that all methods were conducted in compliance with the relevant ethical guidelines and regulations. It has been reviewed and approved by the Academic Committee of the School of Film and Communication at Xiamen University of Technology. Informed consent was obtained from all participants in these studies.

# Result

## Construct validity

The questionnaire items were designed according to existing research to ensure validity and reliability. In translation, a Ph.D. student and two M.A. students conducted a dynamic English-Chinese-English process. Rigorous pretesting, including factor analysis, reliability testing, and expert evaluation in media research, further ensures the applicability of the questionnaire. Based on related studies by Nunnally (1975)[69] and DeVellis and Thorpe (2021)[70] to demonstrate that a Cronbach's α of 0.7 is considered acceptable internal consistency, Hair(2009)[71] further pointed out that when Cronbach's α > 0.7, it is a commonly accepted threshold for judging reliability in social science research, i.e., indicating an acceptable level of scale consistency. The Cronbach's α values for all variables in this study were above 0.7 (see Table 2), indicating that all items measured had good reliability.

| Variable | Refs |
|---|---|
| AI Self Efficacy | 61 |
| Perceived Risks | 62 |
| Personal Innovativeness | 63,64 |
| Perceived Usefulness | 65 |
| Perceived Ease of Use | 66 |
| AI Behavioral Intention | 67,68 |

**Table 1**. Reference for measurement Items.

| Construct | Cronbach's Alpha | KMO |
|---|---|---|
| AI Self Efficacy | 0.871 | 0.925 |
| Perceived Risks | 0.798 | |
| Personal Innovativeness | 0.797 | |
| AI Behavioral Intention | 0.848 | |
| Perceived Usefulness | 0.913 | |
| Perceived Ease of Use | 0.869 | |

**Table 2**. Analysis of reliability and validity of questionnaire.

| Variables | Items | FL | CA | CR | AVE |
|---|---|---|---|---|---|
| AI Self Efficacy | AISE1 | 0.802 | 0.879 | 0.926 | 0.806 |
| | AISE2 | 0.868 | | | |
| | AISE3 | 0.958 | | | |
| Perceived Risks | PR1 | 0.837 | 0.808 | 0.887 | 0.723 |
| | PR2 | 0.851 | | | |
| | PR3 | 0.844 | | | |
| Personal Innovativeness | PI1 | 0.955 | 0.718 | 0.842 | 0.641 |
| | PI2 | 0.767 | | | |
| | PI3 | 0.729 | | | |
| Perceived Usefulness | PU1 | 0.919 | 0.898 | 0.926 | 0.806 |
| | PU2 | 0.926 | | | |
| | PU3 | 0.899 | | | |
| Perceived Ease of Use | PEU1 | 0.885 | 0.865 | 0.917 | 0.787 |
| | PEU2 | 0.894 | | | |
| | PEU3 | 0.899 | | | |
| AI Behavioral Intention | AIBI1 | 0.897 | 0.857 | 0.913 | 0.778 |
| | AIBI2 | 0.901 | | | |
| | AIBI3 | 0.847 | | | |

**Table 3**. Convergent validity assessment.

## Convergent validity

This study employed Smart-PLS to assess convergent validity and internal consistency. As emphasized by Cheung and Cooper-Thomas (2024)[72], relying solely on Cronbach's alpha and factor analysis is inadequate for ensuring measurement quality. In line with Astuti (2021)[73], composite reliability (CR) values in PLS-SEM should exceed 0.70 to meet acceptable standards of internal consistency. Empirical evidence from Cheung and Cooper-Thomas (2024)[72] further supports the convention that CR values greater than 0.70 indicate adequate construct reliability. In this study, CR values range from 0.842 to 0.926, all surpassing the recommended threshold, thereby confirming strong internal consistency across all constructs.

Convergent validity was further assessed using the Average Variance Extracted (AVE). Following the seminal criterion proposed by Fornell and Larcker (1981)[74], AVE values greater than 0.50 indicate that a latent construct explains more than half of the variance in its indicators, thereby demonstrating adequate convergent validity. In the present study, AVE values ranged from 0.641 to 0.806, all exceeding the recommended threshold of 0.50, confirming that the constructs exhibit satisfactory convergent validity (Table 3).

To evaluate multicollinearity, variance inflation factors (VIF) were computed after assessing internal consistency. Following Hair (2009)[75], VIF values should not exceed 10. As shown in Table 4, the VIF ranged from 1.282 to 3.174, confirming the absence of significant multicollinearity and validating the accuracy of the initial model.

Based on the above results, it can be concluded that the structural equation model used in this study shows no significant bias. Further validation of the model's fit was conducted using Smart-PLS, with key PLS-SEM indicators including the standardized root mean square residual (SRMR), normed fit index (NFI), and RMS_theta, as detailed in Table 5. The SRMR value of 0.051 signifies an excellent fit, as it falls well below the commonly accepted threshold for a good model fit. Additionally, as noted by Demler and Paynter et al. (2015)[76], the NFI reflects the degree of correspondence between observed variables and the hypothesized model, with values closer to 1 indicating a better fit. The NFI value of 0.857 in this study falls within the acceptable range, further confirming the model's adequacy.

Additionally, this study used the Fornell-Larcker criterion to assess discriminant validity. As shown in Table 6, the square root of the AVE for each latent construct exceeds the correlations with other constructs, confirming adequate discriminant validity. Beyond the Fornell-Larcker test, the Heterotrait-Monotrait (HTMT) ratio was

| Items | VIF | Items | VIF |
|---|---|---|---|
| AIBI1 | 2.352 | PI1 | 1.503 |
| AIBI2 | 2.45 | PI2 | 1.576 |
| AIBI3 | 1.871 | PI3 | 1.282 |
| PR1 | 1.826 | PU1 | 2.721 |
| PR2 | 1.666 | PU2 | 3.174 |
| PR3 | 1.809 | PU3 | 2.565 |
| PEU1 | 2.271 | AISE1 | 2.337 |
| PEU2 | 2.439 | AISE2 | 2.436 |
| PEU3 | 2.064 | AISE3 | 2.523 |

**Table 4**. Number of variance inflation factor values.

| | Saturated model | Estimated model |
|---|---|---|
| SRMR | 0.051 | 0.071 |
| d_ULS | 0.441 | 0.864 |
| d_G | 0.288 | 0.343 |
| Chi-square | 1053.327 | 1161.946 |
| NFI | 0.857 | 0.842 |

**Table 5**. Structural equation modeling (SEM) statistics.

| | AIBI | PR | PEU | PI | PU | AISE |
|---|---|---|---|---|---|---|
| **AIBI** | 0.882 | | | | | |
| **PR** | 0.457 | 0.850 | | | | |
| **PEU** | 0.627 | 0.377 | 0.887 | | | |
| **PI** | 0.664 | 0.459 | 0.595 | 0.800 | | |
| **PU** | 0.731 | 0.424 | 0.716 | 0.568 | 0.911 | |
| **AISE** | 0.717 | 0.469 | 0.768 | 0.596 | 0.764 | 0.898 |

**Table 6**. Fornell–larcker test for discriminant validity test.

| | AIBI | PR | PEU | PI | PU | AISE |
|---|---|---|---|---|---|---|
| AIBI | | | | | | |
| **PR** | 0.547 | | | | | |
| **PEU** | 0.726 | 0.448 | | | | |
| **PI** | 0.844 | 0.602 | 0.753 | | | |
| **PU** | 0.832 | 0.497 | 0.810 | 0.706 | | |
| **AISE** | 0.824 | 0.556 | 0.878 | 0.748 | 0.859 | |

**Table 7**. HTMT (heterotrait–monotrait ratio of correlations) values.

also employed to further evaluate measurement validity, providing a precise assessment of discriminant validity between constructs.

Discriminant validity in this study was assessed using the Heterotrait–Monotrait (HTMT) ratio as the validation criterion. Following the threshold proposed by Li (2025)[77], HTMT values below 0.90 indicate that the model demonstrates satisfactory discriminant validity. The verification was conducted using Smart-PLS, and 95% confidence intervals were obtained through bootstrapping. The detailed results are presented in Table 7. The HTMT values in this study ranged from 0.448 to 0.878. These results indicate that all latent constructs exhibit adequate discriminant validity, thereby supporting the overall rationality and reliability of the measurement model.

## Predictive power of the model

To evaluate the predictive capability of the structural equation model, this study employs the coefficient of determination ($R^2$), a key indicator of the model's predictive relevance. According to the criteria proposed by

| | $Q^2$ | PLS-SEM_RMSE | PLS-SEM_MAE | LM_RMSE | LM_MAE |
|---|---|---|---|---|---|
| **AIBI1** | 0.511 | 0.51 | 0.364 | 0.513 | 0.369 |
| **AIBI2** | 0.463 | 0.54 | 0.397 | 0.547 | 0.407 |
| **AIBI3** | 0.373 | 0.67 | 0.506 | 0.664 | 0.473 |
| **PEU1** | 0.54 | 0.502 | 0.346 | 0.507 | 0.36 |
| **PEU2** | 0.482 | 0.582 | 0.417 | 0.583 | 0.421 |
| **PEU3** | 0.44 | 0.602 | 0.42 | 0.61 | 0.421 |
| **PU1** | 0.514 | 0.562 | 0.407 | 0.566 | 0.408 |
| **PU2** | 0.503 | 0.553 | 0.397 | 0.558 | 0.398 |
| **PU3** | 0.498 | 0.557 | 0.382 | 0.561 | 0.386 |
| **PU4** | 0.511 | 0.523 | 0.381 | 0.526 | 0.384 |

**Table 8**. Assessing the predictive power of models.

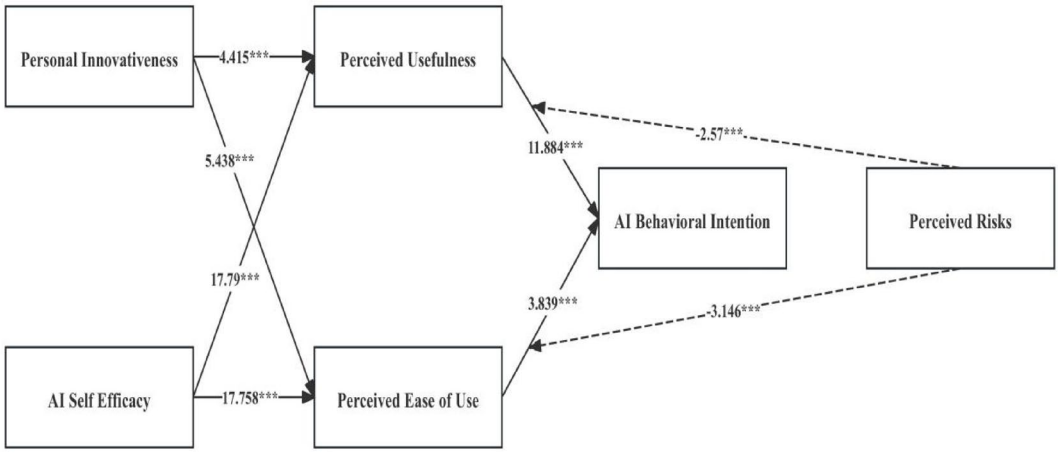


**Fig. 2**. Structural equation path and significance.

Figueiredo Filho et al., the coefficient of determination is calculated as the ratio of the regression sum of squares to the total sum of squares, which indicates the extent to which the predictor variables account for the variance in the dependent variable. In this study, the $R^2$ values range from 0.557 to 0.620.

To validate the research model's predictive power, an out - of - sample prediction assessment was conducted using PLSpredict following Shmueli and Ray et al. (2016)[78]. Table 8 shows $Q^2$ values of all endogenous constructs range from 0.373 to 0.540. According to He and Ren (2025)[79], $Q^2 > 0.35$ means high predictive relevance. For out - of - sample prediction errors, we used Shmueli et al. (2019)[80] classification standards. If PLS - SEM has lower RMSE and MAE than LM and $Q^2 > 0$, it has strong predictive power; if it outperforms LM on part of indicators with $Q^2 > 0$, it has moderate power; if PLS - SEM performs worse than LM and $Q^2 < 0$, it lacks power. In this study, PLS - SEM prediction errors are comparable to or better than LM for most indicators. For items AIBI1–AIBI3, PEU1–PEU2, and PU1–PU4, PLS - SEM RMSE and MAE are lower than LM. Overall, since all $Q^2$ values exceed 0.35 and PLS - SEM outperforms LM on most indicators, the proposed model has predictive ability.

## Path analysis and hypothesis testing

After conducting the above validations, the study applied SmartPLS bootstrap resampling (5000 iterations) to perform hypothesis testing and determine path coefficients, t-values, p-values, and confidence intervals (see Fig. 2). Table 9 summarizes the data analysis results and the hypothesis test outcomes: Personal innovativeness has a significant positive effect on perceived ease of use, with a path coefficient of $p = 0$, $p < 0.001$, standard deviation (SD) = 0.039, and t-value = 5.438. Personal innovativeness also significantly influences perceived usefulness, with a path coefficient of $p = 0$, $p < 0.001$ (SD = 0.04, t = 4.415), indicating a significant positive impact. AI self-efficacy significantly influences perceived ease of use, with a path coefficient of $p = 0$, $p < 0.001$ (SD = 0.036, t = 17.758), demonstrating a strong positive effect. AI self-efficacy also significantly impacts perceived usefulness, with a path coefficient of $p = 0$, $p < 0.001$, SD = 0.037, and a t-value of 17.79. Perceived ease of use significantly affects AI behavioral intention, with a path coefficient of $p = 0$, $p < 0.001$ (SD = 0.05, t = 3.839), indicating a positive effect. Similarly, perceived usefulness significantly influences AI behavioral intention, with a path coefficient of $p = 0$, $p < 0.001$ (SD = 0.044, t = 11.884), confirming a strong positive effect on behavioral intention. Based on the empirical analysis of the data above, it can be concluded that hypotheses H1, H2, H3, H4, H5, and H8 are supported.

| | β | SD | Confidence Interval | | t | *p* | Signaifcan | Decision |
|---|---|---|---|---|---|---|---|---|
| | | | 2.50% | 97.50% | | | | |
| Direct Impact | | | | | | | | |
| PI-PEU | 0.21 | 0.039 | 0.14 | 0.294 | 5.438 | 0 | < 0.05 | Validated |
| PI-PU | 0.177 | 0.04 | 0.099 | 0.255 | 4.415 | 0 | < 0.05 | Validated |
| AISE-PEU | 0.645 | 0.036 | 0.567 | 0.708 | 17.758 | 0 | < 0.05 | Validated |
| AISE-PU | 0.684 | 0.037 | 0.584 | 0.73 | 17.79 | 0 | < 0.05 | Validated |
| PEU-AIBI | 0.196 | 0.05 | 0.19 | 0.93 | 3.839 | 0 | < 0.05 | Validated |
| PU-AIBI | 0.601 | 0.044 | 0.434 | 0.608 | 11.884 | 0 | < 0.05 | Validated |
| Indirect Impact | | | | | | | | |
| PI-PU-AIBI | 0.106 | 0.023 | 0.05 | 0.139 | 4.074 | 0 | < 0.05 | Validated |
| AISE-PU-AIBI | 0.411 | 0.032 | 0.276 | 0.417 | 3.834 | 0 | < 0.05 | Validated |
| PI-PEU-AIBI | 0.041 | 0.014 | 0.017 | 0.073 | 2.828 | 0 | < 0.05 | Validated |
| AISE-PEU-AIBI | 0.126 | 0.032 | 0.06 | 0.185 | 3.834 | 0 | < 0.05 | Validated |
| Overall Impact | | | | | | | | |
| PI-AIBI | 0.147 | 0.027 | 0.082 | 0.188 | 4.944 | 0 | < 0.05 | Validated |
| AISE-AIBI | 0.537 | 0.035 | 0.399 | 0.534 | 13.571 | 0 | < 0.05 | Validated |

**Table 9**. Path coefficients of the research framework.

Additionally, when validating perceived usefulness as a mediator, the results show that the PI-PU-AIBI path has a coefficient of $p = 0$, $p < 0.001$ (SD = 0.023, t = 4.074), and the AISE-PU-AIBI path has a coefficient of $p = 0$, $p < 0.001$ (SD = 0.032, t = 3.834). Similarly, when validating perceived ease of use as a mediator, the PI-PEU-AIBI path has a coefficient of $p = 0$, $p < 0.001$ (SD = 0.014, t = 2.828), and the AISE-PEU-AIBI path has a coefficient of $p = 0$, $p < 0.001$ (SD = 0.032, t = 3.834). Finally, considering the overall effects, the path from personal innovativeness to AI behavioral intention has a coefficient of $p = 0$, $p < 0.001$ (SD = 0.027, t = 4.944), and the path from AI self-efficacy to AI behavioral intention has a coefficient of $p = 0$, $p < 0.001$ (SD = 0.035, t = 13.571). Combining direct and indirect effects, the study confirms that perceived usefulness and perceived ease of use mediate the relationship between personal innovativeness, AI self-efficacy, and AI behavioral intention. Based on these empirical findings, hypotheses H6, H7, H9, and H10 are supported.

## The intrinsic drivers of personal innovativeness and AI self efficacy

Hypotheses H1 and H2 in this study receive strong support, aligning with previous research, particularly in studies integrating TAM and the Task-Technology Fit model. Chen (2019)[81] integrated the TTF model into the TAM framework and similarly confirmed that personal innovativeness significantly and positively influences both perceived usefulness and perceived ease of use. Zehir (2021)[82] further validated that individuals with higher levels of personal innovativeness tend to have more optimistic expectations about the applicability of emerging technologies. These individuals are more likely to focus on technological innovation and demonstrate enhanced abilities to perceive the usefulness and ease of use of intelligent technologies.

Hypotheses H3 and H4 are also supported by the data, further validating previous research findings. Zhang and Yu et al.(2020)[83] similarly incorporated AI self-efficacy into TAM and found that individuals with higher AI self-efficacy were more likely to perceive the usefulness and ease of use of technology, which in turn facilitated their adoption of intelligent technologies. Joo and Park et al.(2018)[84] analyzed the TAM among South Korean university students and discovered that AI self-efficacy significantly and positively influenced perceived usefulness and perceived ease of use. Additionally, students with higher levels of AI self-efficacy tend to exhibit stronger intentions to use the technology in practice. Furthermore, recent studies by Hoang and Ho et al.(2025)[85], Mohamed Eldakar and Khafaga Shehata et al.(2025)[86] have also confirmed a significant positive correlation between students' AI self-efficacy and their perceptions of the usefulness and ease of use of AI technologies during the adoption process.

## Direct effects and the manifestation of influencing factors

Perceived usefulness demonstrates a significant direct positive effect on media students' behavioral intention to use intelligent tools (t = 11.884, $p$ = 0), thereby validating hypothesis H5. This finding illustrates that the evaluation of AI tools' utility directly shapes behavior intention. Specifically, individuals' cognitive assessments, formed through their experiences with AI tools, influence their behavior via an associative mechanism. Similarly, perceived ease of use also significantly and positively influences media students' behavioral intention to use AI tools (t = 3.839, $p$ = 0), providing robust support for hypothesis H8. These results align with the studies of Humbani and Wiese (2019)[87] and Tseng and Hsu et al.(2022)[88], which also found a significant positive correlation between perceived usefulness, perceived ease of use, and behavioral intention of AI tools. This "cognition-behavior" bidirectional framework highlights how AI tools, through both external and internal factors, influence individual cognition, thereby enriching the explanatory mechanism of how perceptions of technological accessibility affect usage behavior.

## Mediation effects and the explanation of pathways

In the "PI-PU-AIBI" path analysis (t = 4.074, $p$ = 0, Confidence Interval (CI) = [0.05, 0.19]), perceived usefulness significantly mediates the relationship between personal innovativeness and AI behavioral intention. The confidence interval, which does not include zero, indicates that this mediating effect is statistically significant and has practical relevance. This suggests that individuals with higher levels of innovation are more likely to perceive the usefulness of AI tools, ultimately enhancing their behavioral intention. Similarly, in the "PI-PEU-AIBI" path (t = 2.828, $p$ = 0, CI = [0.017, 0.073]), perceived ease of use also plays a significant mediating role in the relationship between personal innovativeness and AI behavioral intention. Again, the confidence interval does not include zero, confirming both statistical significance and practical relevance of this mediating effect. Therefore, hypotheses H6 and H9 are supported, offering a detailed explanation of how an individual's internal characteristics, such as innovation capacity, influence actual usage outcomes through the perception of AI tools' usefulness. Celik and Ayaz(2025)[89] similarly emphasize that perceived usefulness and perceived ease of use serve as mediating pathways between personal innovativeness and AI behavioral intention.

In the "AISE-PU-AIBI" path (t = 3.848, $p$ = 0, confidence interval [CI] = [0.276, 0.417]), the study demonstrates that AI self-efficacy enhances AI behavioral intention by increasing perceived usefulness when using intelligent tools. The confidence interval excludes zero, indicating that this mediating effect is statistically significant and substantively meaningful. Similarly, in the "AISE-PEU-AIBI" path (t = 3.848, $p$ = 0, CI = [0.032, 0.06]), perceived ease of use significantly mediates the effect of AI self-efficacy on AI behavioral intention; the confidence interval excludes zero, confirming the statistical significance and practical relevance of this mediation. These findings provide empirical support for hypotheses H7 and H10, elucidating how AI self-efficacy as an intrinsic characteristic influences the behavioral intention of intelligent tools. Consistent with this, Thüs and Malone et al. (2024)[90], Mun and Hwang (2025)[91] found that perceived usefulness and perceived ease of use significantly mediate the relationship between AI self-efficacy and AI behavioral intention among media students.

## Moderating effects revealing the influence mechanism

The moderating effects were examined using the PROCESS macro in SPSS 29. Latent construct scores for perceived usefulness, perceived ease of use, and perceived risk were generated using factor score weights from confirmatory factor analysis (CFA), a common two - stage approach for individual - level latent variable estimates for moderation testing (Pieters et al., 2022)[92]. Before computing interaction terms, all predictor and moderator variables were mean - centered to reduce multicollinearity and stabilize estimates (Frazier et al., 2004)[93]. Then, interaction terms were created by multiplying the centered scores of perceived risk with perceived usefulness and perceived ease of use respectively, following standard moderated regression analysis procedures. Simple - slope analyzes were conducted and visualized in Excel, with p - values formatted to three decimal places and slope plots showing interaction effects at ± 1 standard deviation of the moderator. These procedures ensure rigorous moderation analysis and accurate interpretation of the negative moderating role of perceived risk on the relationships between perceived usefulness, perceived ease of use, and AI behavioral intention.

Path 1(See Figure 3) corresponds to the interaction effect of perceived risk and perceived usefulness on AI behavioral intention, with values at one standard deviation below the mean (M − 1SD = 3.518) and above the mean (M + 1SD = 3.591). The effect size difference at M + 1SD is 0.073, with t = − 2.57 and $p$ = 0.001, indicating that perceived risk significantly and negatively moderates the relationship between perceived usefulness and AI behavioral intention.

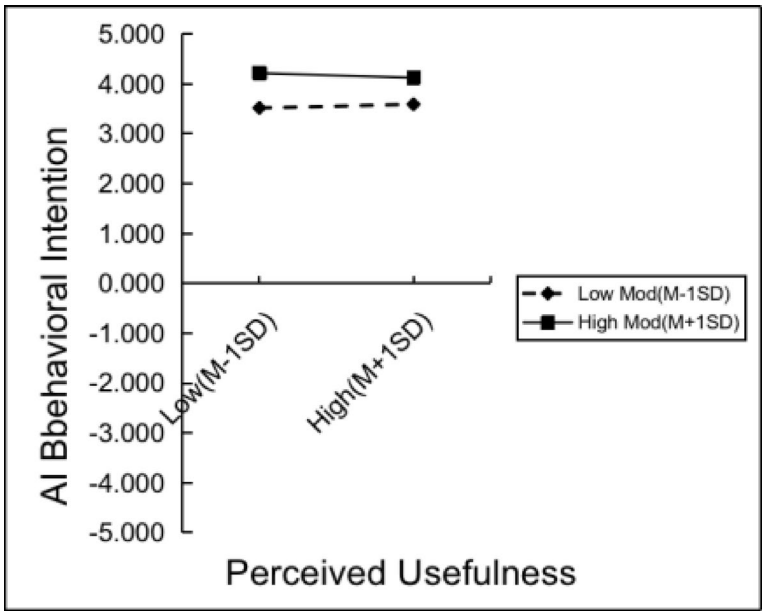


**Fig. 3**. PRxPU-AIBI simple slope plot.

Path 1(See Figure 4)corresponds to the interaction effect of perceived risk and perceived ease of use on AI behavioral intention, with values at M − 1SD = 3.453 and M + 1SD = 3.687. The effect size difference at M ± 1SD is 0.234, with t = − 3.146 and $p = 0.002$, suggesting a significant negative moderating effect of perceived risk on the relationship between perceived ease of use and AI behavioral intention (Fig. 3).

The hypotheses H11 and H12 of this study are supported by the data. Pioneering research by Ye and Yan et al. (2024)[94], Yang and Ding et al. (2024)[95] also indicates that perceived risk negatively influences individuals' ability to perceive the usefulness and ease of use of technology, thereby negatively mediating the relationship between perceived usefulness, perceived ease of use, and behavioral intention (Fig. 4).

## FsQCA

*Variable calibration*

Before conducting the fsQCA, the standardized latent construct scores derived from the PLS-SEM results were used as inputs for calibration. Following Ragin's (2009)[96] direct method, raw values were transformed into fuzzy-set membership scores ranging from 0 to 1. Consistent with standard practice in configurational research (Shah and Corley, 2006[97]; Schneider & Wagemann, 2012[98], the 95th, 50th, and 5th percentiles of each variable were selected as the thresholds for full membership, the crossover point, and full non-membership, respectively. The 95/50/5 calibration scheme has become one of the most widely adopted heuristics in configurational studies because it maximizes discrimination across membership levels, ensures clear separation between full membership and non-membership, and minimizes artificial clustering around the point of maximum ambiguity (Woodside,2013)[99].

To evaluate the robustness of these calibration thresholds, an alternative scheme using the 90th, 50th, and 10th percentiles was additionally tested. This alternative is frequently employed in fsQCA studies when researchers seek to reduce the influence of extreme values in small-to-medium samples or when the upper tail of the distribution exhibits limited dispersion (Misangyi et al., 2017)[100]. The solution pathways and truth tables generated under the 90/50/10 thresholds were substantively identical to those obtained under the 95/50/5 scheme, indicating that the findings are robust to reasonable variations in calibration anchors. Furthermore, in accordance with Ragin's (2009)[101] recommendation, all membership scores equal to 0.5 were increased by 0.001 to avoid the potential exclusion of cases positioned precisely at the point of maximum ambiguity.

*Necessity analysis*

Following data calibration, a necessity analysis was conducted within the fsQCA framework. Before proceeding to the sufficiency analysis, it is essential to verify that the consistency scores for all antecedent conditions remain below the threshold of 0.90. As recommended by Ragin's (2009)[101] and further formalized by Crilly and Zollo et al., (2012)[102], a consistency value of 0.90 or higher indicates a necessary condition for a causal relationship. In contrast, values below this benchmark suggest that no single antecedent exerts a deterministic effect. Therefore, when the consistency level is less than 0.9, the configuration effect analysis can be carried out.

In this study, all antecedent conditions exhibited consistency scores lower than 0.90 (see Table 10), indicating that none of them qualify as necessary conditions influencing individuals' intention to use AI tools. Consequently, it is appropriate to investigate further how these conditions interact and operate in combination to shape media students' behavioral intentions to engage with AI technologies. The findings underscore the multidimensional and complex nature of behavioral intention toward AI usage, where no single factor functions

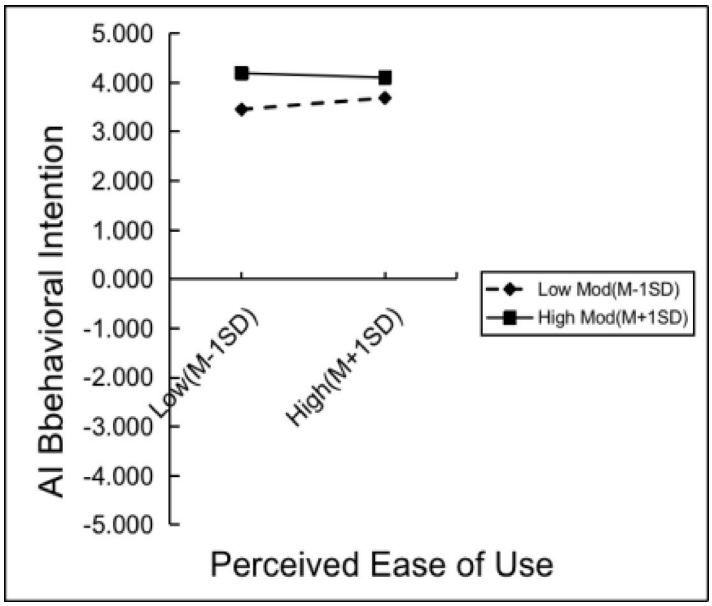


**Fig. 4**. PRxPEU-AIBI simple slope plot.

| AI Behavioral Intention | | | ~AI Behavioral Intention | |
|---|---|---|---|---|
| **Items** | **Consistency** | **Coverage** | **Consistency** | **Coverage** |
| **AISE** | 0.846375 | 0.816381 | 0.495508 | 0.618618 |
| **~AISE** | 0.604605 | 0.480769 | 0.852922 | 0.87784 |
| **PI** | 0.843064 | 0.820016 | 0.483112 | 0.608207 |
| **~PI** | 0.597194 | 0.471638 | 0.857034 | 0.876059 |
| **PR** | 0.792841 | 0.723558 | 0.549447 | 0.649015 |
| **~PR** | 0.615407 | 0.513453 | 0.765968 | 0.827162 |
| **PU** | 0.65672 | 0.584958 | 0.613408 | 0.707189 |
| **~PU** | 0.671267 | 0.572929 | 0.639997 | 0.707009 |
| **PEU** | 0.604683 | 0.625418 | 0.55295 | 0.740236 |
| **~PEU** | 0.748848 | 0.564116 | 0.720191 | 0.702205 |

**Table 10**. Analysis of necessary conditions based on fsQCA.

| | High AI Behavioral Intention | | |
|---|---|---|---|
| **Condition Variables** | **H1** | **H2** | **H3** |
| PI | ● | ● | |
| AISE | ● | ⊗ | ● |
| PU | • | • | ⊗ |
| PEU | ⊗ | ⊗ | ⊗ |
| PR | ⊗ | ⊗ | • |
| Raw coverage | 0.537155 | 0.398432 | 0.490835 |
| Unique coverage | 0.197185 | 0.0463989 | 0.115859 |
| Consistency | 0.876777 | 0.856454 | 0.886698 |
| Solution coverage | 0.77518 | | |
| Solution consistency | 0.845182 | | |

**Table 11**. Configurations of high AI behavioral intention for users. Note: A solid large circle(●)indicates the presence of a core condition, while a solid small circle(•)denotes the presence of a peripheral condition. A crossed large circle (⊗) signifies the absence of a core condition, and a crossed small circle (⊗) indicates the absence of a peripheral condition. A blank cell represents a "don't care" condition, meaning that the presence or absence of this factor has no impact on the given configuration pathway.

as a decisive determinant in isolation. Therefore, developing effective mechanisms to promote the adoption of AI tools requires a comprehensive and integrative approach.

*Sufficiency analysis*

Based on the necessity analysis, this study conducted a sufficiency analysis of five antecedent variables, following the established standards from previous research. The original consistency threshold was set at 0.8, and the frequency threshold was set at 1. In line with prior studies, the analysis primarily focused on the antecedent variables that appeared in both the parsimonious and intermediate solutions as core conditions, while those that only appeared in the intermediate solution were treated as peripheral conditions. The core conditions identified indicate a strong association with the outcome variable, while variables appearing solely as peripheral conditions suggest a weaker relationship with the outcome. Through multiple analyzes, three configuration paths were identified to explain the behavioral intention of media students in China to use AI tools. The overall coverage of these three paths was 0.77518, and the overall consistency was 0.84518, indicating that these configurations can explain approximately 77.5% of the cases (see Table 11).

The overall consistency and coverage values obtained in this study fall within the generally accepted and widely applied ranges for fsQCA research. According to the methodological guidelines established by Rihoux and Ragin (2009)[103], an overall consistency level above 0.80 indicates that the configurational solutions reliably represent a subset relationship. The overall coverage (0.77518) and consistency (0.84518) observed in the present study align well with established methodological benchmarks and fall squarely within the representative intervals reported in high-impact configurational research.

Configuration S1 (PU*PI*SE) represents a driving pathway composed of two core conditions—personal innovativeness and AI self-efficacy—together with perceived usefulness as a peripheral condition. Personal innovativeness and AI self-efficacy construct strong endogenous personal traits, creating a solid psychological foundation for individuals' behavioral intention of intelligent technologies. Perceived usefulness, as an external influencing factor, further enhances students' positive perceptions of AI tools, which is particularly significant in

| Condition Variables | High Continuance Intention | | |
|---|---|---|---|
| | H1 | H2 | H3 |
| PI | ● | ● | ⊗ |
| AISE | ● | ⊗ | ● |
| PU | • | ● | ⊗ |
| PEU | ⊗ | ⊗ | ⊗ |
| PCR | ⊗ | ⊗ | • |
| Raw coverage | 0.646116 | 0.576971 | 0.566051 |
| Unique coverage | 0.0453345 | 0.021209 | 0.015965 |
| Consistency | 0.846416 | 0.850717 | 0.879787 |
| Solution coverage | 0.842749 | | |
| Solution consistency | 0.807876 | | |

**Table 12**. Path of robustness test Results. Note: A solid large circle (●) indicates the presence of a core condition, while a solid small circle (•) denotes the presence of a peripheral condition. A crossed large circle (⊗) signifies the absence of a core condition, and a crossed small circle (⊗) indicates the absence of a peripheral condition. A blank cell represents a "don't care" condition, meaning that the presence or absence of this factor has no impact on the given configuration pathway.

the context of higher media education. In this configuration, the absence of perceived ease of use and perceived risk does not exert a significant impact on individuals' AI behavioral intention. Thus, the dual endogenous drivers of personal innovativeness and AI self-efficacy constitute a powerful self-motivation mechanism.

Configuration S2 (PI*~SE*PU*~PEU*~PR) identifies personal innovativeness as a core condition and perceived usefulness as a peripheral condition, while AI self-efficacy is a core absent condition and perceived risk, Perceived ease of use, a peripheral absent condition. This suggests that, in the absence of concerns about AI-related risks, personal innovativeness becomes a decisive factor influencing media students' actual usage of intelligent technologies. When individuals demonstrate strong innovative thinking and are free from anxiety or fear about technological risks and algorithmic bias, they are able to effectively perceive the usefulness of these technologies. Consequently, innovativeness emerges as the primary driving force behind the adoption of AI tools in this configuration.

Configuration S3 (~PU*~PEU*PR*SE) identifies AI self-efficacy as a core condition and perceived risk as a peripheral condition, while perceived usefulness and perceived ease of use are classified as peripheral absent conditions. This configuration indicates that when perceived risk is high, it undermines individuals' perceptions of the usefulness and ease of use of AI tools, thereby diminishing external advantages. In such circumstances, AI self-efficacy plays a decisive role: individuals with high levels of AI self-efficacy are able to effectively resist the influence of perceived risk, and their acceptance and application of intelligent technologies are driven primarily by their internal psychological traits. Notably, the presence of perceived risk indeed exerts a negative effect on perceptions of usefulness and ease of use, weakening the influence of these peripheral conditions; however, the impact of AI self-efficacy as an intrinsic motivational driver becomes correspondingly more salient.

*Robustness test*

We further conducted robustness tests on the results: (1) Case impact: re-evaluation indicates that there is no case-driven solution, and the necessary conditions have not been fulfilled. (2) Solution type: the complex and intermediate solutions of the research results remain unchanged, and the application of these solutions aligns with the anticipated conclusions. (3) Threshold stringency: by increasing the original consistency threshold from 0.80 to 0.85 during the reconstruction of the truth table and adjusting the PRI consistency threshold accordingly, the study found that the core configuration path remains stable. Therefore, these analyzes demonstrate that the fsQCA analysis of this study produces a solution path that is both robust and reliable (Table 12).

## Discussion

### Findings and theoretical contributions

This study clarifies the mechanisms and pathways influencing media students' behavioral intention to use AI tools in higher education. By adopting a mixed-methods approach that extends the traditional TAM and integrates PLS-SEM with fsQCA, the research develops innovative validation strategies for intelligent media education. The findings yield three major theoretical contributions: (1) Identifying the factors that influence Chinese media students' adoption of intelligent tools; (2) Incorporating individual internal characteristics into the classical TAM framework, thereby extending the traditional model with differentiated mechanisms; and (3) Combining quantitative analysis through PLS-SEM with fsQCA to jointly validate and enhance the pathways that promote intelligent tool adoption among Chinese media students.

First, the results from PLS-SEM analysis show that personal innovativeness, as a significant antecedent of individual internal characteristics, not only has a significant impact on perceived usefulness (H1) but also positively affects perceived ease of use (H2). This study extends the traditional framework of TAM, clarifying the critical role of individual internal characteristics in facilitating the perception and implementation of technology during the acceptance process. The findings further demonstrate that media students' perceptions

of AI technology's usefulness and ease of use are largely driven by their own innovativeness. Individuals with higher innovativeness tend to be more curious and eager about emerging smart technologies, showing increased attention. Compared to average individuals, they are more likely to seek early adoption and experience with new technologies (Price and Ottati et al.,2015)[104]. This innovation-driven priority is consistent with the findings of Jahanmir and Silva et al. (2020)[105], which emphasize that personal innovativeness is a core factor in enhancing individuals' perceived usefulness, ease of use, and behavioral intentions in intelligent environments.

The findings also demonstrate that AI self-efficacy significantly and positively influences both perceived usefulness (H3) and perceived ease of use (H4). Unlike previous studies that examined intelligent technology acceptance solely through the traditional TAM framework (Chatterjee and Rana,2021)[106], this study incorporates AI self-efficacy as an antecedent variable within the model. The results further indicate that AI self-efficacy enhances individuals' behavioral intention by improving their work efficiency and task management capabilities. Individuals with higher AI self-efficacy believe they can effectively handle AI-related tasks with minimal external support and possess the skills to utilize diverse intelligent technologies. Consequently, this strengthens their perceptions of usefulness and ease of use in practical work settings (Jiang et al., 2021)[107].

Secondly, regarding the classic TAM constructs, perceived usefulness and perceived ease of use, the results firstly demonstrate that PU exerts a significant effect on AI behavioral intention (H5). Moreover, PU functions as a mediating variable, strengthening the impact of antecedents such as personal innovativeness and AI self-efficacy on AI behavioral intention (H9 and H11). These findings further corroborate prior research: Alyoussef and Drwish et al.(2025)[108] argued that when AI tools effectively solve practical problems, users develop strong perceptions of usefulness, which in turn significantly influence their AI behavioral intention. Uddin and Thiam-Low et al. (2024)[109] found a significant positive correlation between students' perceived usefulness and actual behavioral intention when integrating AI tools in higher education. Additionally, students with higher levels of innovativeness tend to perceive greater usefulness of the technology. Alyoussef and Drwish et al.(2025)[108] also highlighted that AI tools characterized by user-friendliness and high functional value are more likely to elicit strong perceived usefulness during user experience, thereby enhancing behavioral intention.

The results validated by PLS-SEM further demonstrate that perceived ease of use has a significant positive effect on AI behavioral intention (H6). Additionally, it is evident that perceived ease of use also mediates the influence of personal innovativeness and AI self-efficacy on AI behavioral intention, playing a significant positive role (H10 and H12). Abdullah and Ward et al.(2016)[110] found that the introduction of AI technologies into art education significantly enhanced students' perceived ease of use, thereby increasing their AI behavioral intention. This trend is expected to become more widespread in future smart art education initiatives. Furthermore, perceived ease of use plays a significant mediating role between personal innovativeness, AI self-efficacy, and actual behavioral intention (Xu and Zhang et al., 2023)[111]. Sánchez-Prieto and Cruz-Benito et al.,(2020)[112]further emphasized the relationship between students' perception of the ease of use of AI technologies and their intention to use intelligent tools in smart education, demonstrating that perceived ease of use positively predicts individuals' AI behavioral intention.

Finally, the structural equation modeling results reveal a complex interplay among perceived risk, perceived usefulness, perceived ease of use, and AI behavioral intention, highlighting a negative moderating mechanism. Users' perceived risks associated with intelligent tools negatively mediates the relationship between perceived usefulness, perceived ease of use, and AI behavioral intention, exerting an inhibitory effect (H7 and H8). Prior research has established perceived risk as a critical determinant in individual decision-making, negatively mediating the association between perceived usefulness, perceived ease of use, and AI behavioral intention. When users experience anxiety or distrust toward intelligent tools during task execution, they may reduce actual use and lower their intention for continued engagement, even if the tools demonstrate utility and ease of use (Hansen and Saridakis et al.,2018)[113]. Man and Xiong et al. (2020)[114] note that generative AI operates as a "black box" while consuming vast amounts of personal data, with opaque internal processes infringing on data owners' rights. Moreover, ambiguous data policies and unclear privacy terms exacerbate users' concerns over data risks, ultimately causing even those with strong perceived usefulness and ease of use to abandon AI technologies due to apprehensions about data and algorithmic risks. Therefore, it is crucial to integrate AI algorithmic mechanisms and ethical frameworks into media curricula, fostering ongoing reflection and refinement of AI algorithm design, while providing a clear understanding of the trade-offs between AI-related risks and rewards.

In summary, this study constructs a dual-dimensional framework—comprising individual intrinsic traits and external perceptions—to explain the determinants of AI technology adoption among Chinese media students. Based on PLS-SEM path analysis, it demonstrates that PI and AISE serve as core endogenous antecedents driving students' acceptance of intelligent technologies. Meanwhile, PU and PEU, as external cognitive factors, not only significantly enhance AIBI but also mediate the relationship between intrinsic drivers and actual usage. Furthermore, the inclusion of PR reveals the complexity underlying students' decision-making regarding AI adoption in the context of "intelligent+" media education, thereby enriching the theoretical understanding of the cognition–behavior interaction in AI-mediated media education environments.

## Practical implications

This study further employed fsQCA in combination with PLS-SEM to form a mixed-methods approach, thereby offering a novel methodological perspective for examining the factors influencing media students' intentions to adopt AI in their learning. At the same time, this mixed-methods approach provides practical solutions and enhancement strategies for the integration and application of intelligent tools within media.

*Focus on the motivation of media students in higher education*

The findings regarding the intrinsic motivating effects of PI and SE on media students' AIBI indicate that the acceptance and application of AI technologies require attention not only to technological advancements and

external environmental factors but also to individuals' intrinsic motivation. Furthermore, the first configuration path (S1) derived from the fsQCA analysis provides compelling evidence that PI and AISE are key drivers of AI acceptance. This underscores the importance of prioritizing individual willingness rather than direct technological implementation in intelligent media education. Prior to integrating AI technologies into media curricula, universities should systematically assess students' psychological traits, innovative thinking, and sense of efficacy. With a clear understanding of students' adoption intentions, targeted efforts can be made to effectively incorporate intelligent technologies into talent cultivation.

Media education outcomes have long been closely linked to students' individual characteristics (Jaakkola, 2022)[115]. Morris and Yeoman et al.,(2023)[116]have suggested that media education should enhance students' learning motivation and acceptance intentions through more personalized and engaging educational approaches. These strategies remain central to the current integration of media education and intelligent technology development. Nichols and LeBlanc (2021)[117]emphasized that the failure to meet expectations in media education is not due to the curriculum or teaching methods themselves, but rather the limitations in how students and their engagement with the course and instruction are guided and understood—especially the neglect of individuals' intrinsic agency. It is crucial to reevaluate the central role of students in media education, and to effectively foster students' proactive acceptance and active use of emerging technologies by enhancing their innovation awareness and AI self-efficacy.

*Enhancing the creative design and practical operability of media courses*

In the acceptance and application of AI technology, students' perceptions of usefulness and ease of use significantly influence their actual behavioral intentions. Moreover, PU and PEU act as key mediators and moderators of the effects of PI and AISE on AIBI. Furthermore, the fsQCA's second pathway configuration (S2) confirms that the combination of PU and PI substantially enhances AIBI. These findings highlight that, in designing intelligent media education systems, emphasis should be placed on both course novelty and engagement, as well as the practicality and clarity of knowledge delivery, thereby optimizing curriculum content and teaching methods.

According to the learning objectives established by UNESCO for AI courses, the framework for media education should embrace a diverse range of technical interests and applications. A common approach has been to integrate AI-related technologies and projects across various levels of media education, spanning disciplines such as the humanities, arts, language, communication, and sociology, often through interdisciplinary curriculum design (Mohebzadeh and Emamjomeh et al., 2020)[118]. Wang and Liu (2021)[119] further emphasize that, within the context of media education, the primary areas of focus for learners concerning AI technologies are "collaborative tasks," "functional features," and "modeling features."Future strategies for developing content in intelligent media education should prioritize curriculum planning and design within specific technological application scenarios. This approach should also take into account students' individual interests and areas of focus, thereby facilitating personalized course development. For example, courses could be tailored around topics such as intelligent recommendation systems, data-driven content creation, and public opinion analysis, enabling students to understand the relationship between technology and its practical applications. In addition, differentiated module designs should be adopted, offering varied content based on students' individual interests. Examples of such courses include data journalism, intelligent IP design, AI ethics and responsibility, algorithmic governance, and emerging media technologies—courses that cater to students' curiosity and desire to explore the intersection of AI and media. Moreover, interactive and engaging applications should be promoted, including the encouragement of cross-disciplinary collaboration in real-world settings, thereby enhancing students' problem-solving capabilities.

*Deepening professional knowledge and enhancing resilience to risks*

PR associated with AI significantly moderated negatively the relationships between PU, PEU, and AIBI. Furthermore, the third configuration path (S3) identified in the fsQCA analysis indicates that PR impacts behavioral intention. This negative influence underscores the necessity for educational stakeholders to critically examine the comprehensive effects of AI integration within the media industry, acknowledging inherent technological risks and potential threats. Rational reflection and the development of critical thinking and analytical skills are essential for a thorough, bidirectional reconsideration of AI-related educational theories, curriculum development, and pedagogical approaches. The goal is to balance learning demands with technological hazards, thereby fostering resilience to risks and sustaining learners' ongoing willingness to engage. To leverage AI as an enabler of professional development for media students, it is imperative to deepen their understanding of intelligent tools' operational mechanisms, regulatory standards, and ethical implications. Enhancing professional, in-depth, and detailed knowledge of intelligent technologies equips students to better manage potential risks, ultimately strengthening their AI self-efficacy in using AI tools.

In the curriculum design for smart media, the fundamental dimensions encompass the tool attributes and essential characteristics of AI-enabled media in the context of the media discipline. The key themes should focus on fostering an understanding of AI technology's "intrinsic characteristics," "potential risks," and "scope of application." The objective is to equip media students with core knowledge and skills in areas such as intelligent search applications, fake news detection, multimodal content generation, personalized intelligent customization, and digitalized communication. This will enhance their ability to navigate the opportunities and challenges presented by the AI era (Lopezosa and Pont-Sorribes et al., 2023)[120]. Moreover, ethical considerations and a rigorous professional mindset must be integrated into the curriculum to ensure that students maintain confidence and efficacy in their belief that AI can never fully replace human expertise in the media profession.

## Conclusion

This study employs PLS-SEM and fsQCA methods to explore the intrinsic mechanisms and enhancement pathways influencing the acceptance and usage of AI tools among college students majoring in media studies. The PLS-SEM results show that PI, AISE, PU, and PEU all have significant positive effects on AIBI. Consistent with prior research, PU and PEU mediate the relationships between PI, AISE, and AIBI. Additionally, PR negatively mediates the relationships among PU, PEU, and AIBI. The fsQCA analysis further reveals that no single antecedent variable alone constitutes a necessary condition for AIBI among media students; instead, the intention results from the combined effects of multiple factors. Finally, three configurational pathways leading to high behavioral intention are identified: "self-driven," "efficacy-oriented," and "risk-resistant." This study not only identifies the key factors influencing AI adoption among media students but also proposes targeted strategies to enhance adoption rates. The findings underscore that the sustainable integration of AI in media education requires a multifaceted approach, including diversified applications, human-centered practices, and personalized educational strategies. By adopting an individual student perspective, this research provides valuable theoretical insights and practical recommendations for the deep, sustained, and efficient integration of AI technologies into higher education in media studies. These contributions not only enhance the understanding of AI adoption in this field but also offer actionable educational frameworks essential for promoting the long-term application and development of AI in media education.

## Limitations and future research

Due to practical constraints such as limited human resources, time, and material conditions, this study has the following limitations:

(1) The generalizability of the sample remains limited, as the study focuses solely on media students from universities in Fujian Province, China, thereby restricting the scope of inference. Enhancing the representativeness of the sample is essential to improve the wider applicability of the findings.(2) Due to practical constraints, this study did not adopt a longitudinal design and thus did not fully account for the variability of individual characteristics over time. Considering that individual traits may evolve with prolonged exposure and learning of intelligent technologies, the need for long-term, controlled experimental studies merits further investigation. (3) Compared with other structural equation modeling approaches, the PLS-SEM method possesses relatively weaker inferential power and does not deeply explore the causal mechanisms between antecedent factors and actual behavioral intention. Therefore, the explanatory power of the current findings regarding the enhancement of media students' actual use of AI technologies is somewhat limited.

Based on the limitations identified in this study, the following recommendations for future research are proposed: (1) Expand the sample size and incorporate survey data from universities across different regions and include graduate students to enhance the credibility, persuasiveness, generalizability, and representativeness of the findings; (2) Combine questionnaire surveys with controlled experiments, supplemented by both longitudinal and cross-sectional research designs, to ensure more robust and theoretically grounded conclusions; (3) Employ experimental designs and multiple structural equation modeling techniques for deeper validation and supplementary analysis of the data, thereby more accurately revealing the causal relationships among individual traits, external factors, and actual behavioral intentions.

These improvements will provide a stronger theoretical and empirical foundation to support the practical application and sustainable development of AI technologies in media education.

## Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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## Author contributions
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## Declarations

## Competing interests
The authors declare no competing interests.

## Additional information
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